

FRONT ELEVATION WITH CONTEXT

Apertures



SECTION PERSPECTIVE

TO SEPARATE THE SPACES—BEDROOMS AND BATHROOMS—I CREATED PODS CARVED OUT OF THE WALLS. TO ALLOW MORE NATURAL LIGHT, I ADDED OPENINGS THROUGH THE WALLS, ENHANCING THE LIGHTING ON THE FIRST FLOOR AND LIVING SPACES, AND IMPROVING THE VIEWS FROM THE BEDROOMS.

KEYFRAMES





MULTIGENERATIONAL HOUSING

AS CHINESE IMMIGRANTS BEGAN ARRIVING IN PITTSBURGH FROM TAIWAN-CANTON REGION, CHINATOWN TOOK SHAPE IN THE SMALL BUT BUSTLING NEIGHBORHOOD, TRANSFORMING INTO A HOT SPOT OF COMMERCE.

"MORE THAN 100 YEARS LATER, THE NEIGHBORHOOD HAS VANISHED. NOW THE SEAT OF CITY AND COUNTY GOVERNMENT AND HOME TO MANY FORTUNE 500 COMPANIES, AREA RESIDENTS HAVE DISBURSED TO OTHER PARTS OF THE CITY AND OTHER REGIONS OF THE COUNTRY."



MOM: 40 YEARS OLD
FIRST GENERATION IMMIGRANT
MAIN MANAGER OF THE LAUNDROMAT



GRANDMA: 70 YEARS OLD
TAIWAN-CANTON REGION



DAD: 44 YEARS OLD
FIRST GENERATION IMMIGRANT
HUNGRY PANDA DELIVERY GUY



GRANDPA: 77 YEARS OLD
GUANGZHOU CHINA



DAUGHTER: 12 YEARS OLD
SECOND GENERATION IMMIGRANT
MIDDLE SCHOOLER WITH AN IMAGINARY DRAGON FRIEND

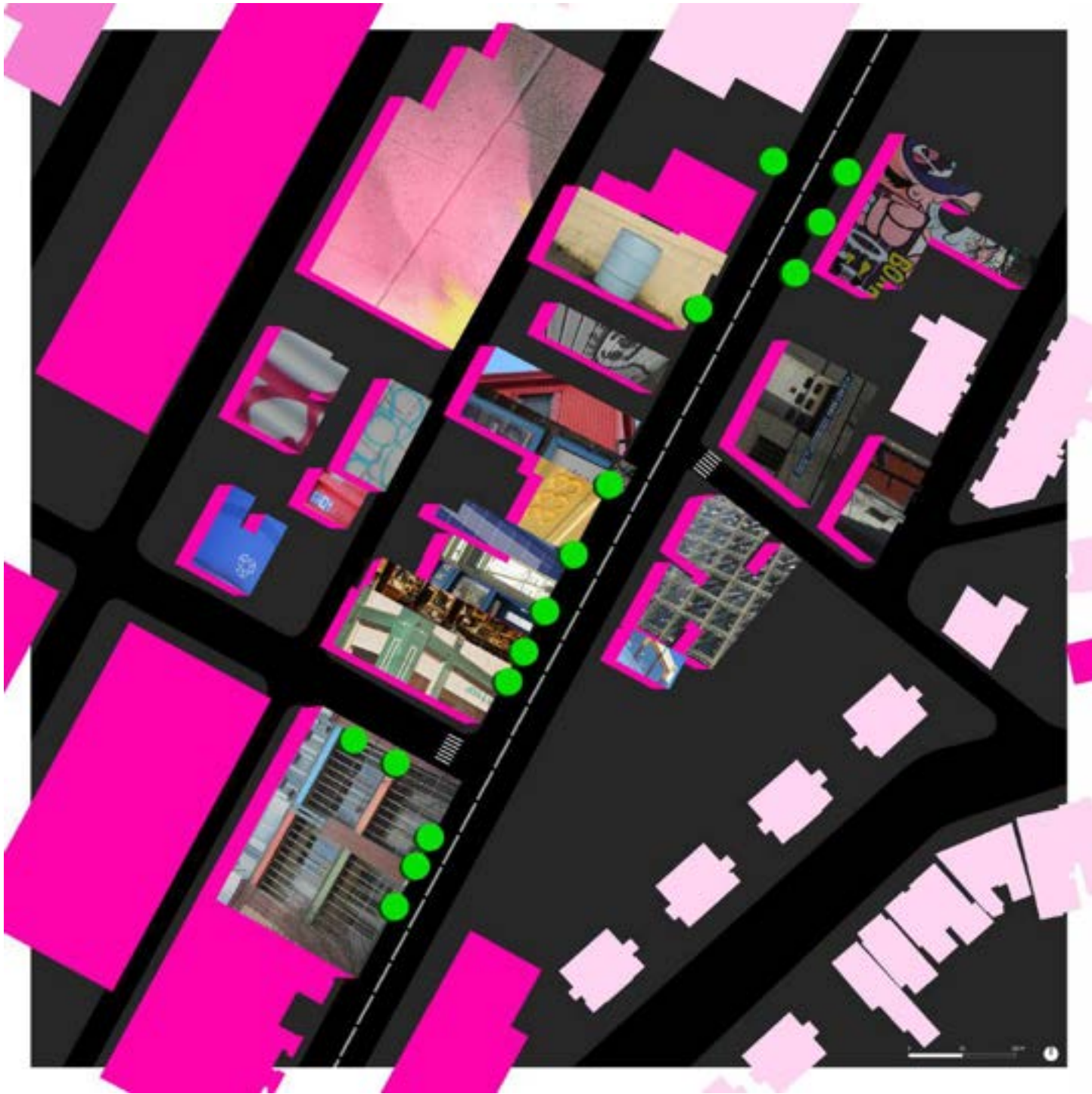


SITE VISIT

LOWER LAWRENCEVILLE IN PITTSBURGH IS A DYNAMIC NEIGHBORHOOD KNOWN FOR ITS BLEND OF HISTORIC CHARM AND MODERN VITALITY. THIS AREA BOASTS A RICH TAPESTRY OF ART, CUISINE, AND BOUTIQUE SHOPPING, MAKING IT A THRIVING HUB FOR BOTH RESIDENTS AND VISITORS SEEKING A UNIQUE, CULTURALLY RICH URBAN EXPERIENCE.

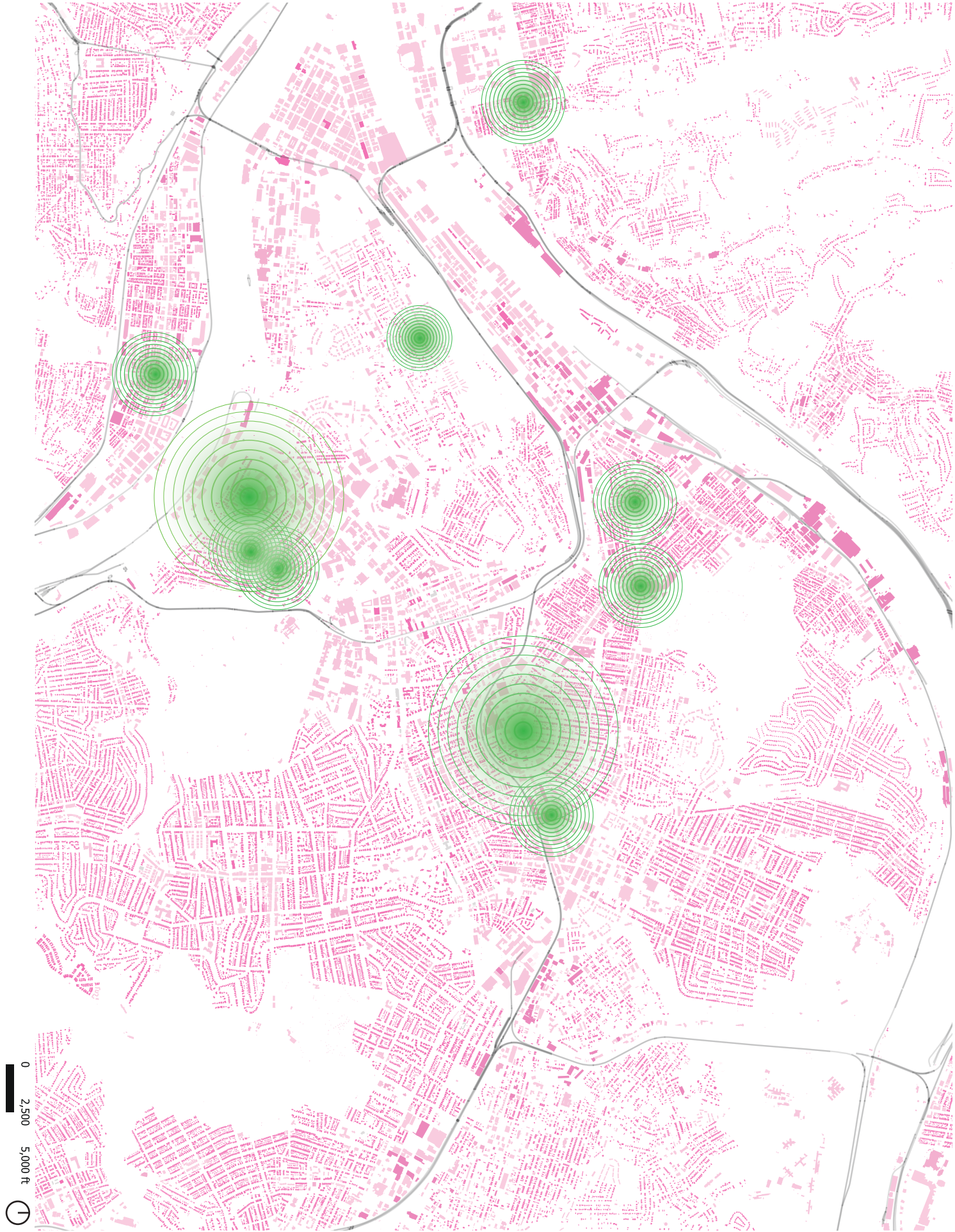
A MAJOR DISTINCTION I FOUND BETWEEN THE NEIGHBORHOOD AND OTHER AREAS IN PITTSBURGH IS THE USE OF VIBRANT/HIGH SATURATION COLORS ON THE STOREFRONTS.

THUS, I DECIDED TO RECORD THE COLORS THAT ARE EYE CATCHING ON THE BLOCK AND LATER TRANSFORMED THEM INTO A SITE MAP THAT DENOTES THE LOCATIONS OF EACH COLOR/PHOTO.



SITE MAP - STOREFRONT

AS MENTIONED BEFORE, I NOTICED THAT THE LOWER LAWRENCEVILLE COMMUNITY IS VERY VIBRANT BOTH IN THE COLOR OF THE BUILDINGS AND THE PEOPLE. THUS FOR THE SITE MAP, I DECIDED TO RECORD THE MOST SIGNIFICANT OBJECT/COLOR OF EVERY STORE THAT SITS BESIDE MY SITE. TO ADD IN ANOTHER LAYER OF INFORMATION, I DECIDED TO COLOR CODE THE NEARBY STRUCTURES BASED ON THEIR FUNCTION WHICH ALSO INFERS THE LIVELINESS OF EACH BUILDING.





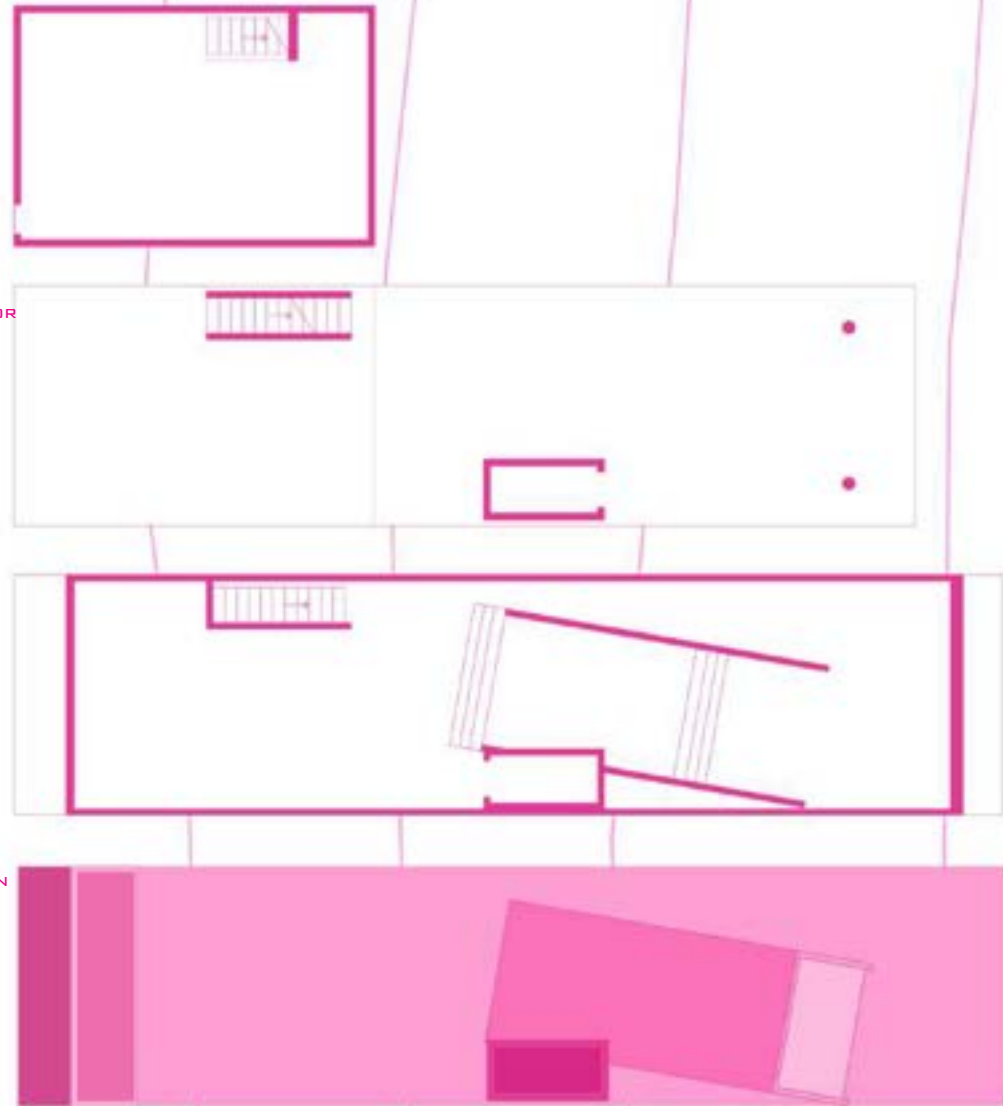
FIRST ITERATIONS - DESIGN 1

LOWER LEVEL

GROUND FLOOR

FIRST FLOOR

TOP ELEVATION

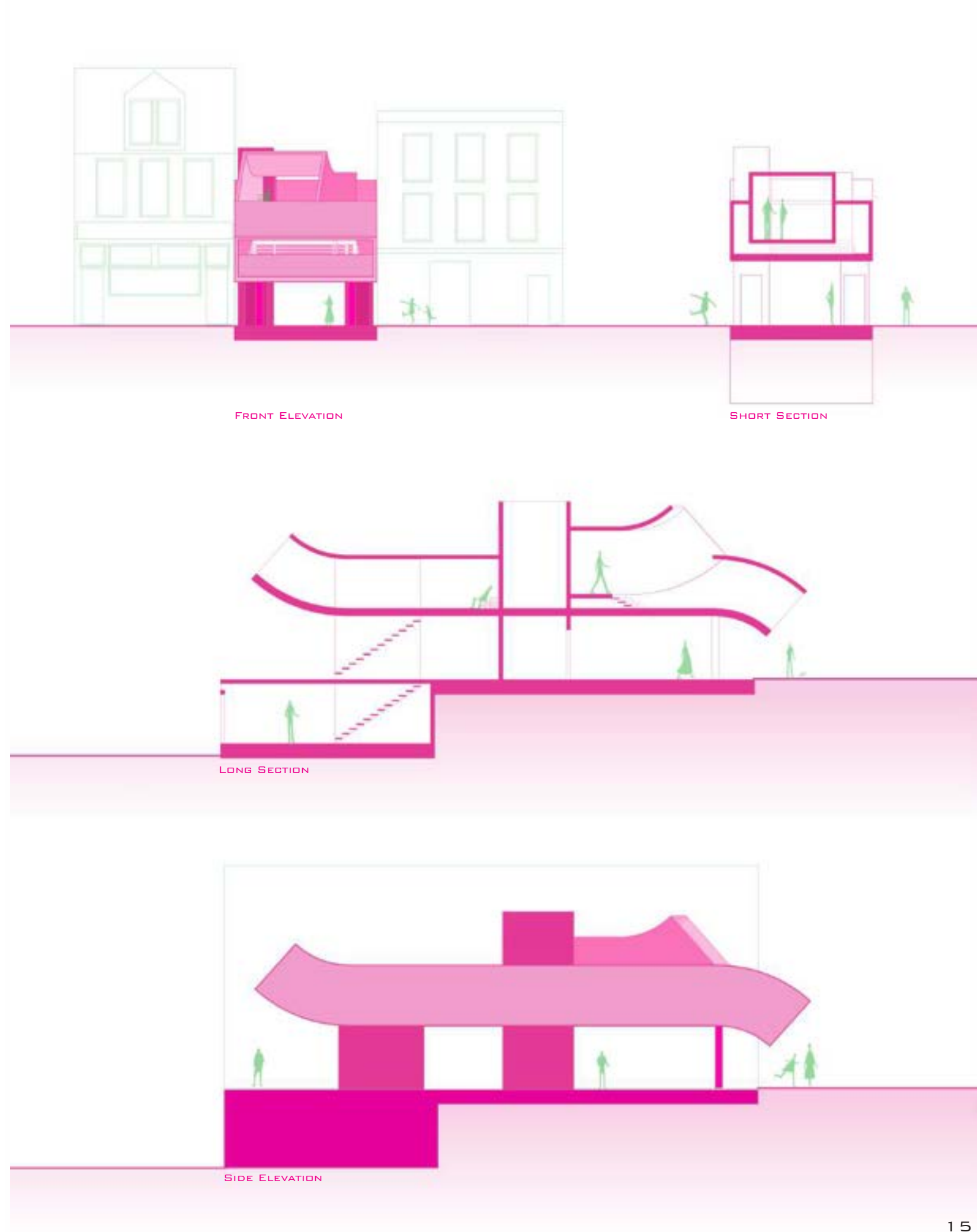


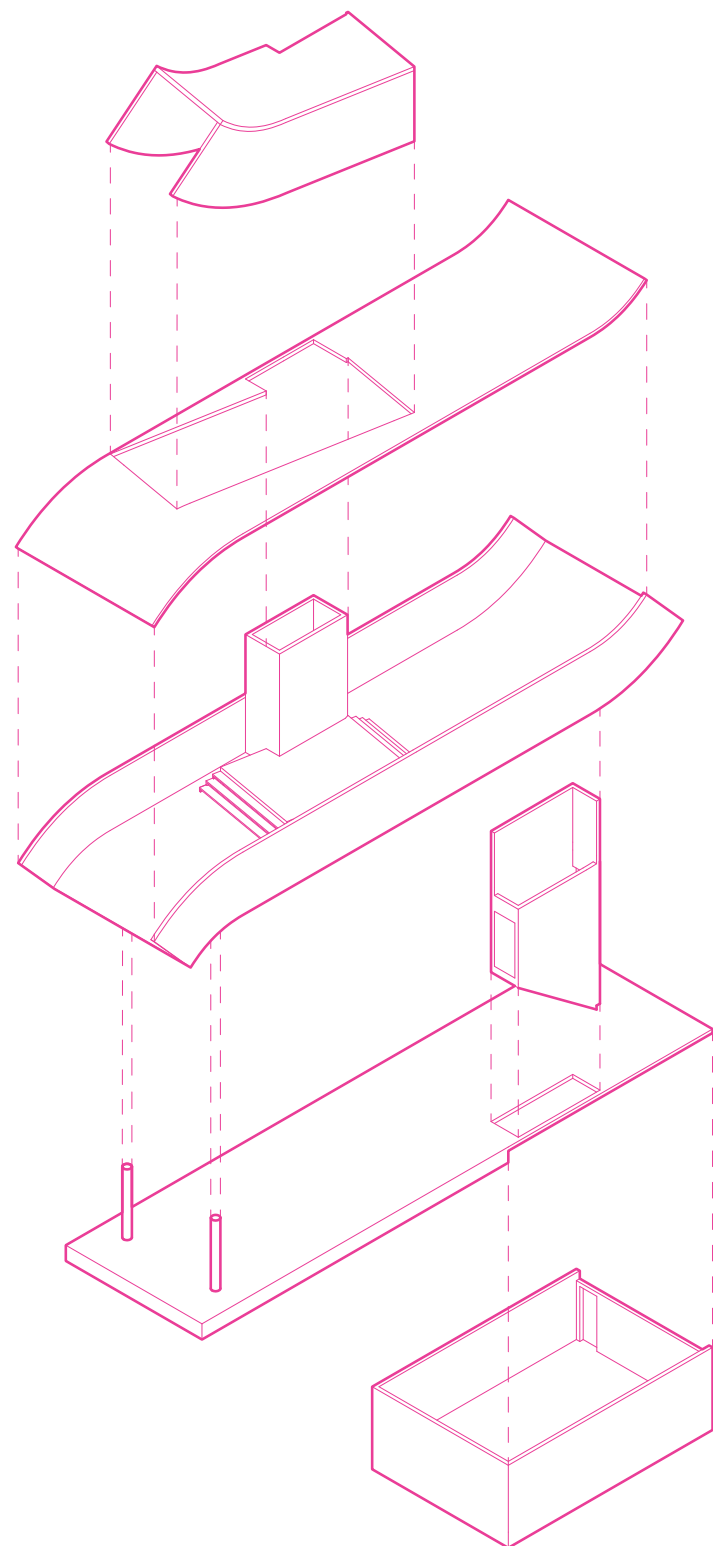
FRONT ELEVATION

SHORT SECTION

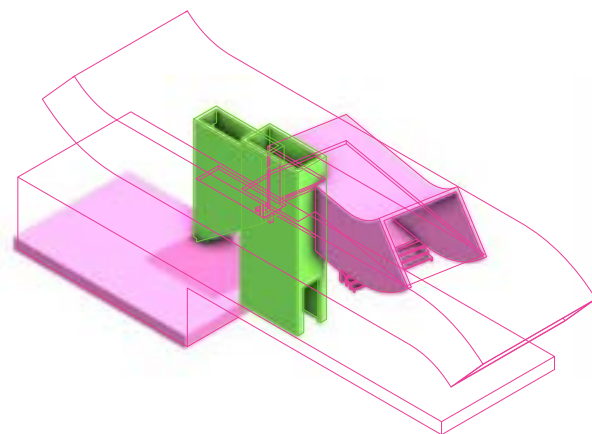
LONG SECTION

SIDE ELEVATION

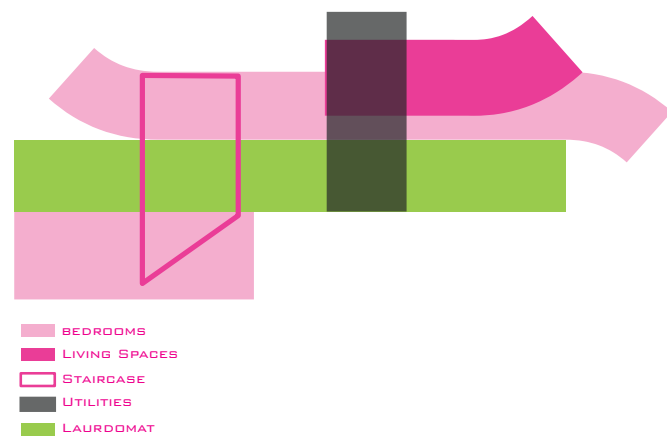




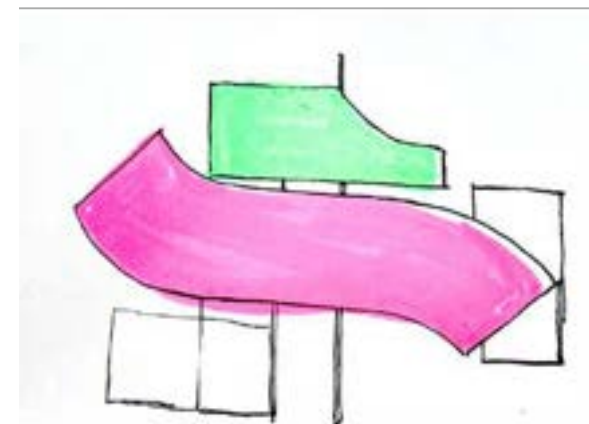
EXPLODED AXON



FORM



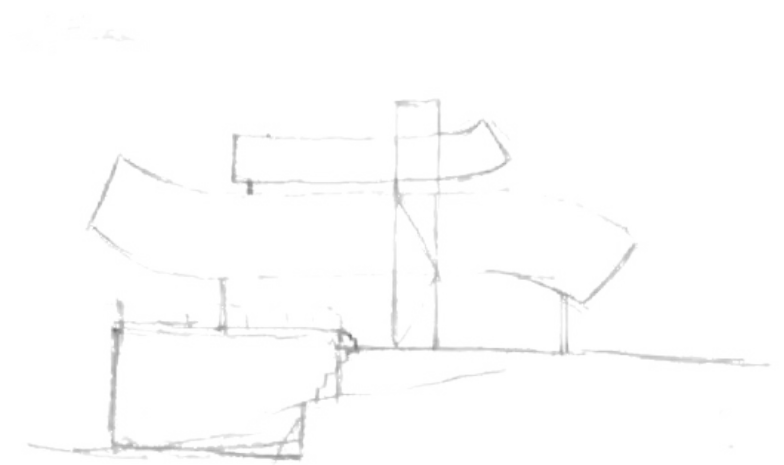
- BEDROOMS
- LIVING SPACES
- STAIRCASE
- UTILITIES
- LAURDOMAT



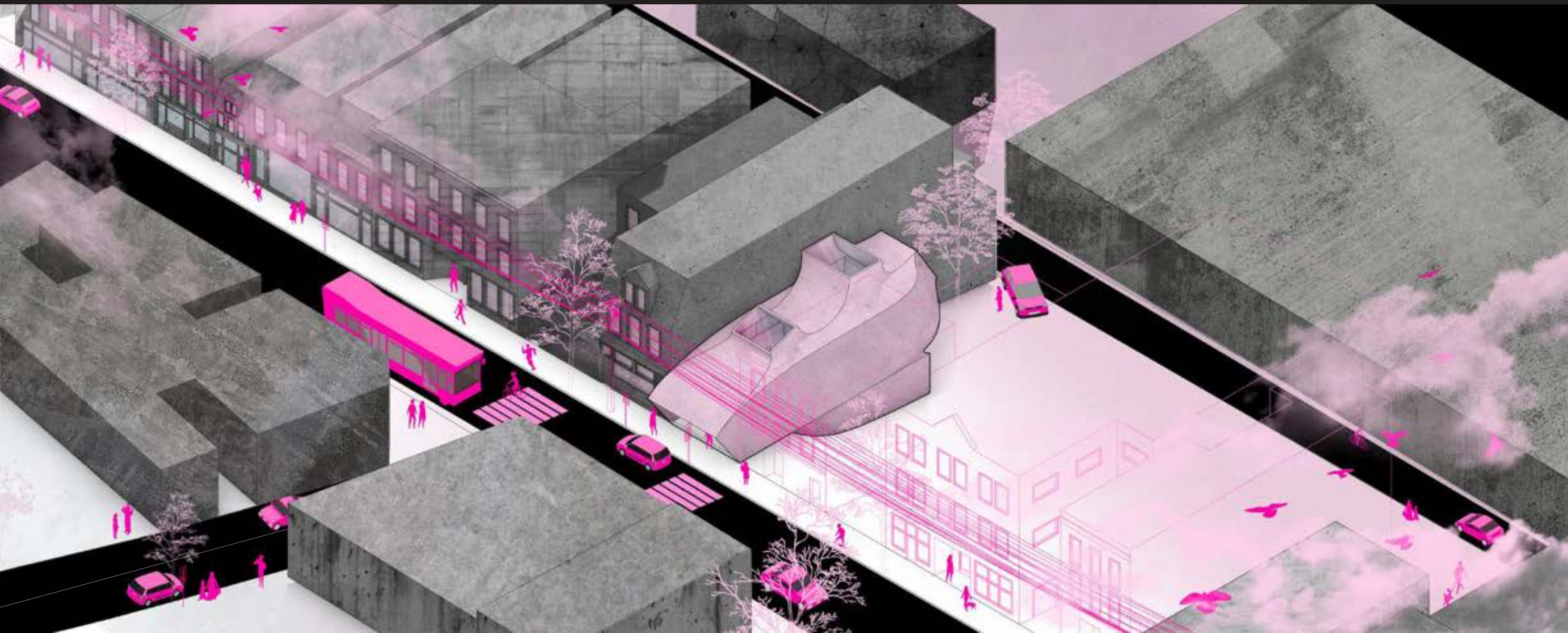
IDEA DEVELOPMENT

THIS IS THE INITIAL PARTI IDEA FOR THE LAURDOMAT WHERE THE THREE FLOORS ARE CONNECTED THROUGH PLANES. THE CURVATURES ARE MEANT TO DIRECT PEOPLE'S VIEWS TOWARDS THE SKY, THE STREETS AND THE HILL.

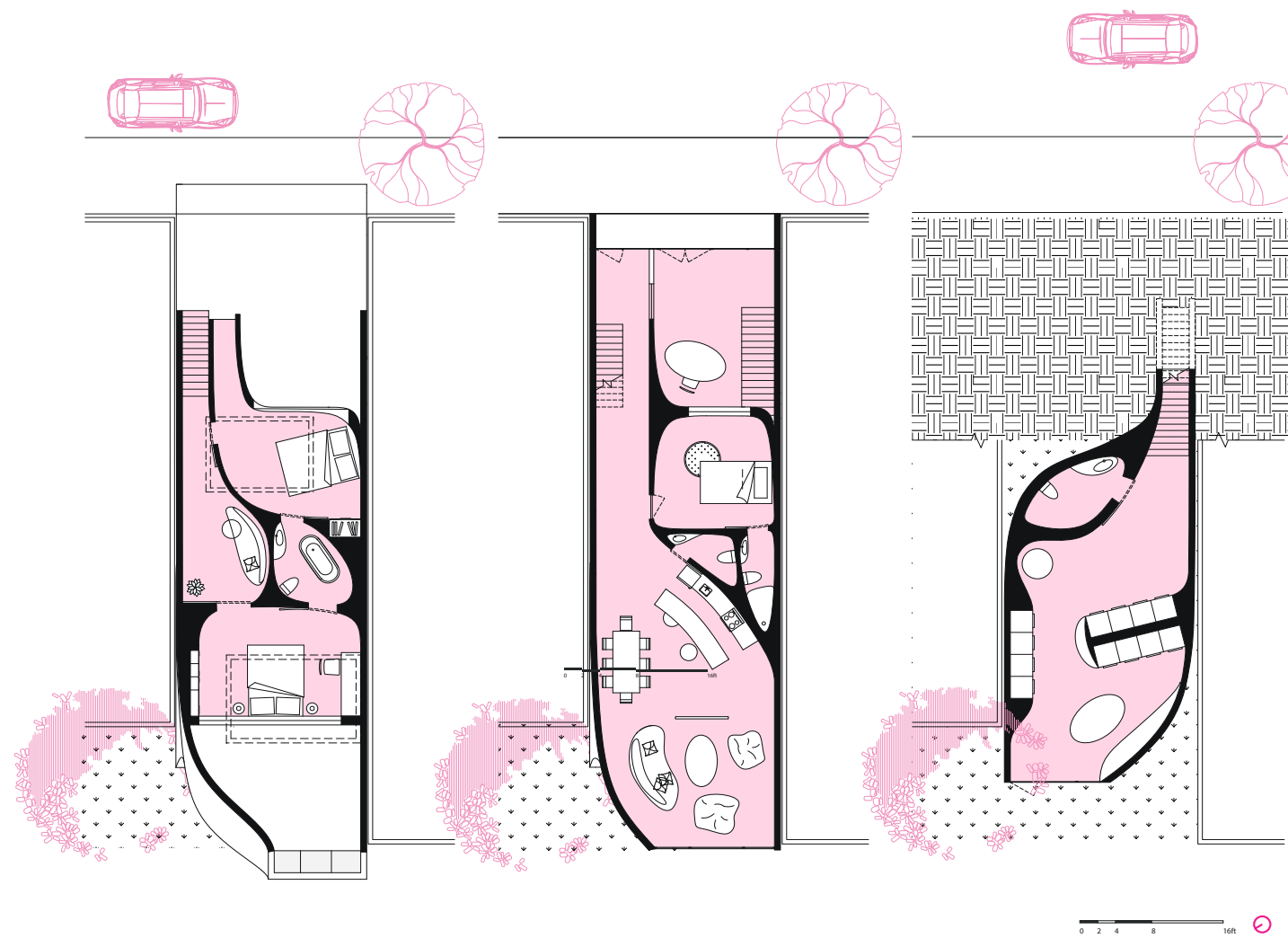
FOR THE SECOND ITERATION, INSTEAD OF HAVING PLANES INTRUDE THE THREE GEOMETRIES, I USED VOLUMES TO CONNECT THE THREE FLOORS AND THE CONVERSION OF PLANES TO VOLUMES ALLOWS MORE VARIETY IN FUNCTIONAL SPACES



FOR FUTURE ITERATIONS, I PLAN TO MAKE THE STRUCTURE A CONTINUOUS GEOMETRY THAT BRANCHES OUT AND SPLITS TO ALLOW CIRCULATIONS.

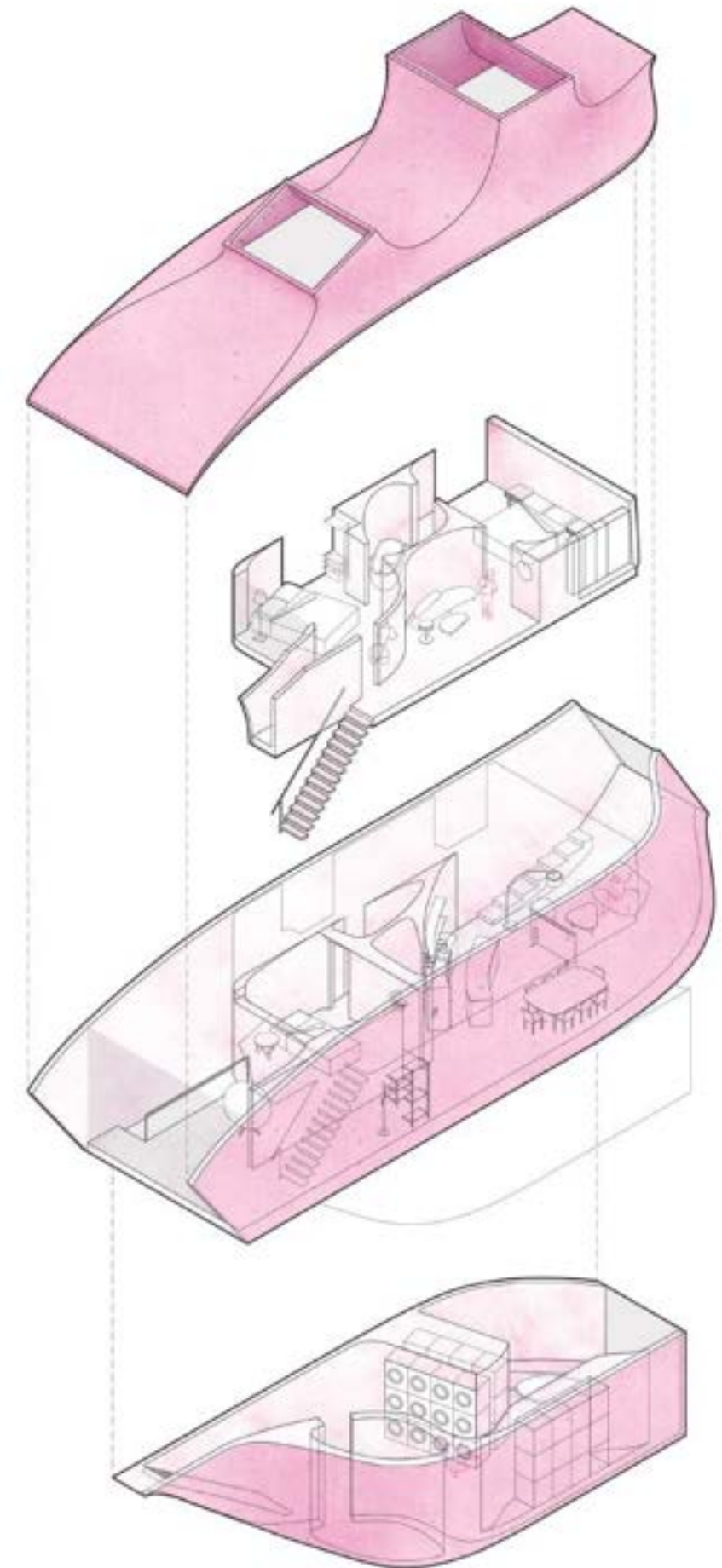


Apertures final Iteration

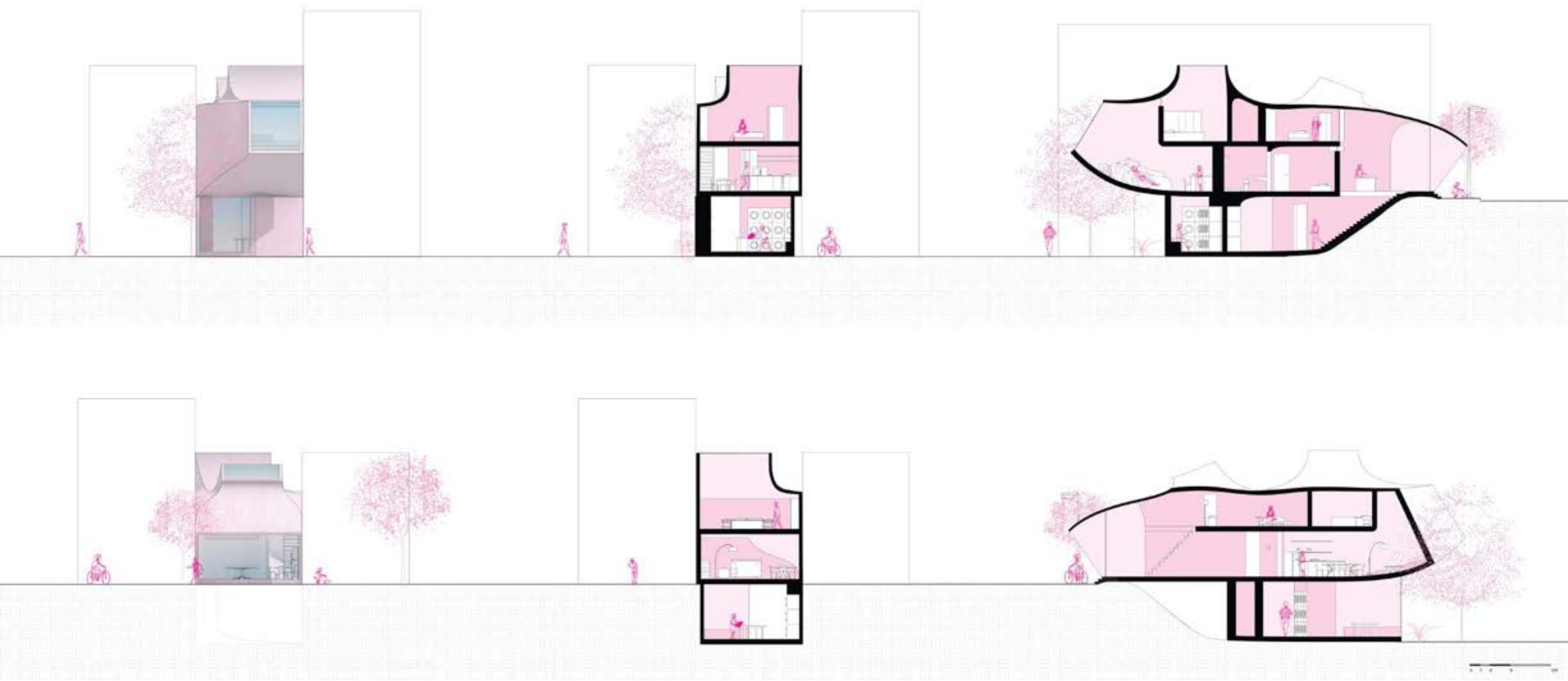


PLANS

ORIGINALLY, THE CURVES EXISTED ONLY IN THE SECTIONS. LATER, I DECIDED TO INCORPORATE THESE CURVES INTO THE PLANS TO GUIDE CIRCULATION. REGARDING PROGRAM ORGANIZATION, I PLACED THE LAUNDROMAT UNDERGROUND WITH FULL ACCESS TO THE ALLEYWAY, WHILE THE COUNTER IS LOCATED ON THE GROUND FLOOR TO ENSURE IT IS WELCOMING ON BUTLER STREET.



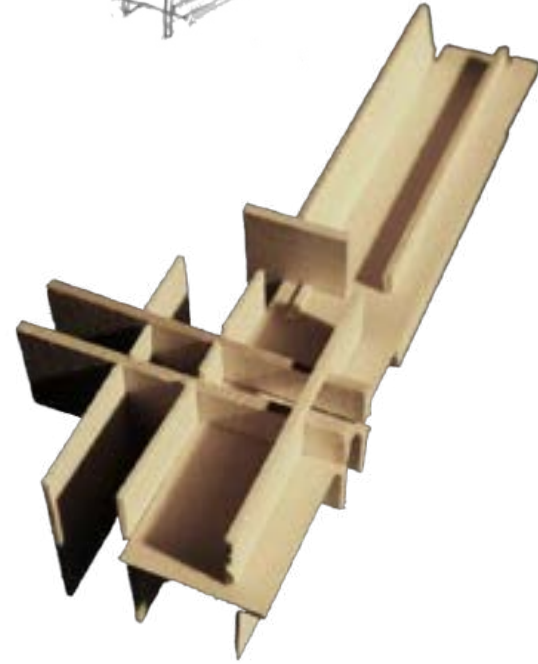
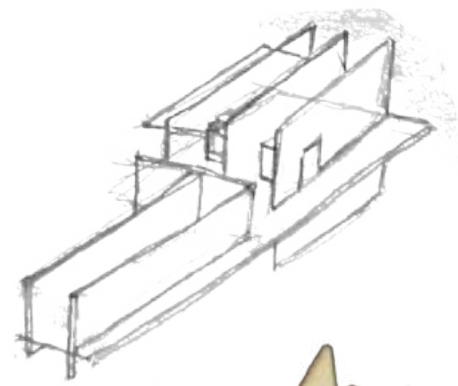
EXPLODED AXONOMETRIC



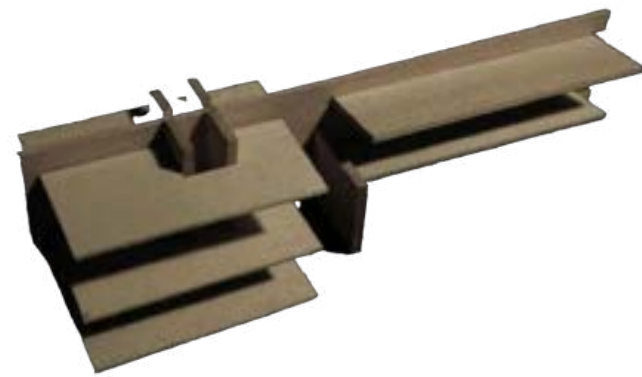
SECTIONS AND ELEVATIONS

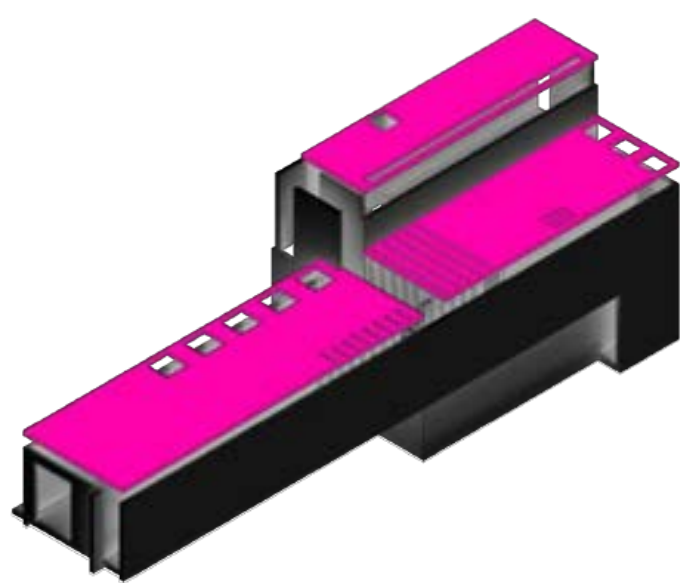
IDEA DEVELOPMENT

THIS IS AN EARLY STAGE SKETCH OF THE MASSING MODEL FOR DESIGN 2. THE SKETCH WAS THEN TRANSLATED INTO AN 1/16" MASSING MODEL AS ONE OF THE FIRST ITERATIONS.

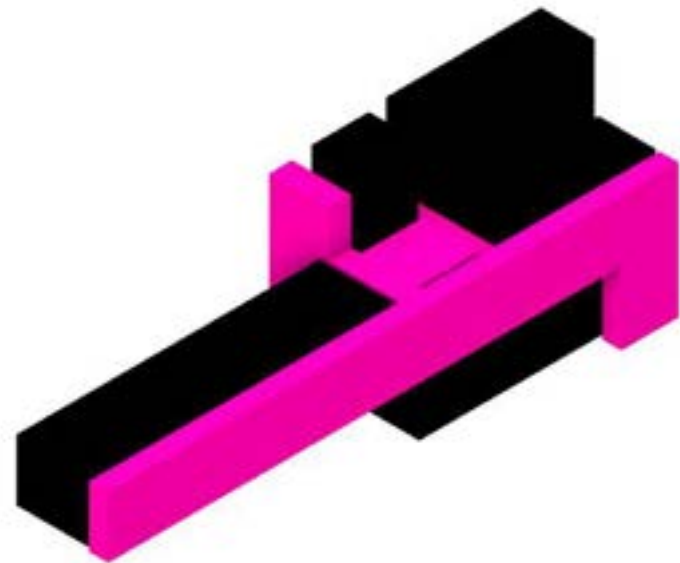


FINAL MASSING MODEL



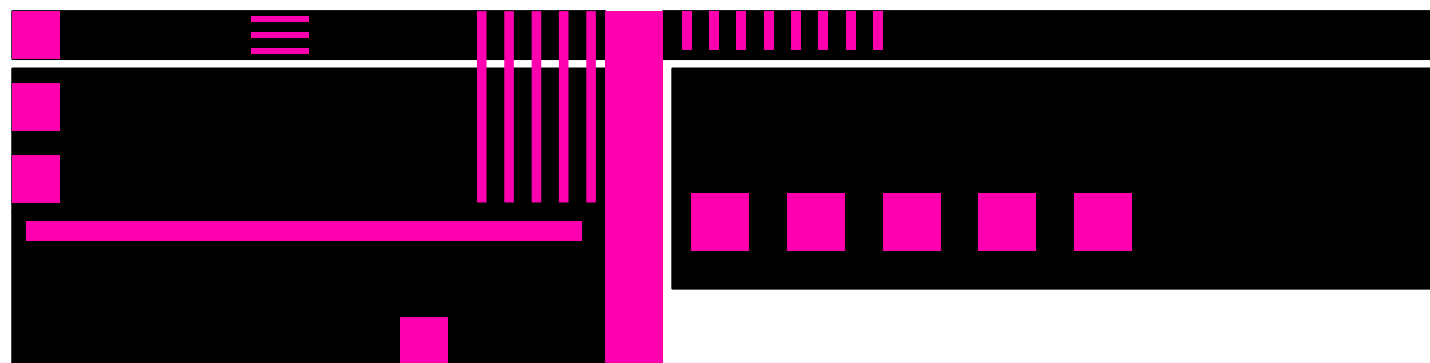


DESIGN 2 AXONOMETRIC



PROGRAM

LIGHTING



 DIRECT SUNLIGHT

 SHADOWS

House In Itami Intervention

Michelle Yi Xuan Guo
Tommy CheeMou Yang
Jared Abraham
Jerry(Yonghao) Zhang



Intervention Model Side Elevation(Right)

PROJECT INTRODUCTION

The precedent for this project is the House in Itami, which locates in Itami, Japan, covering an area of around 300 square feet. The intervention aims to reorganize the circulation, considering that my client, a seamstress, will need to move large objects around the house. Furthermore, the extensions of the new system towards the sides also redefines space by intruding into the existing structure.





Background Information

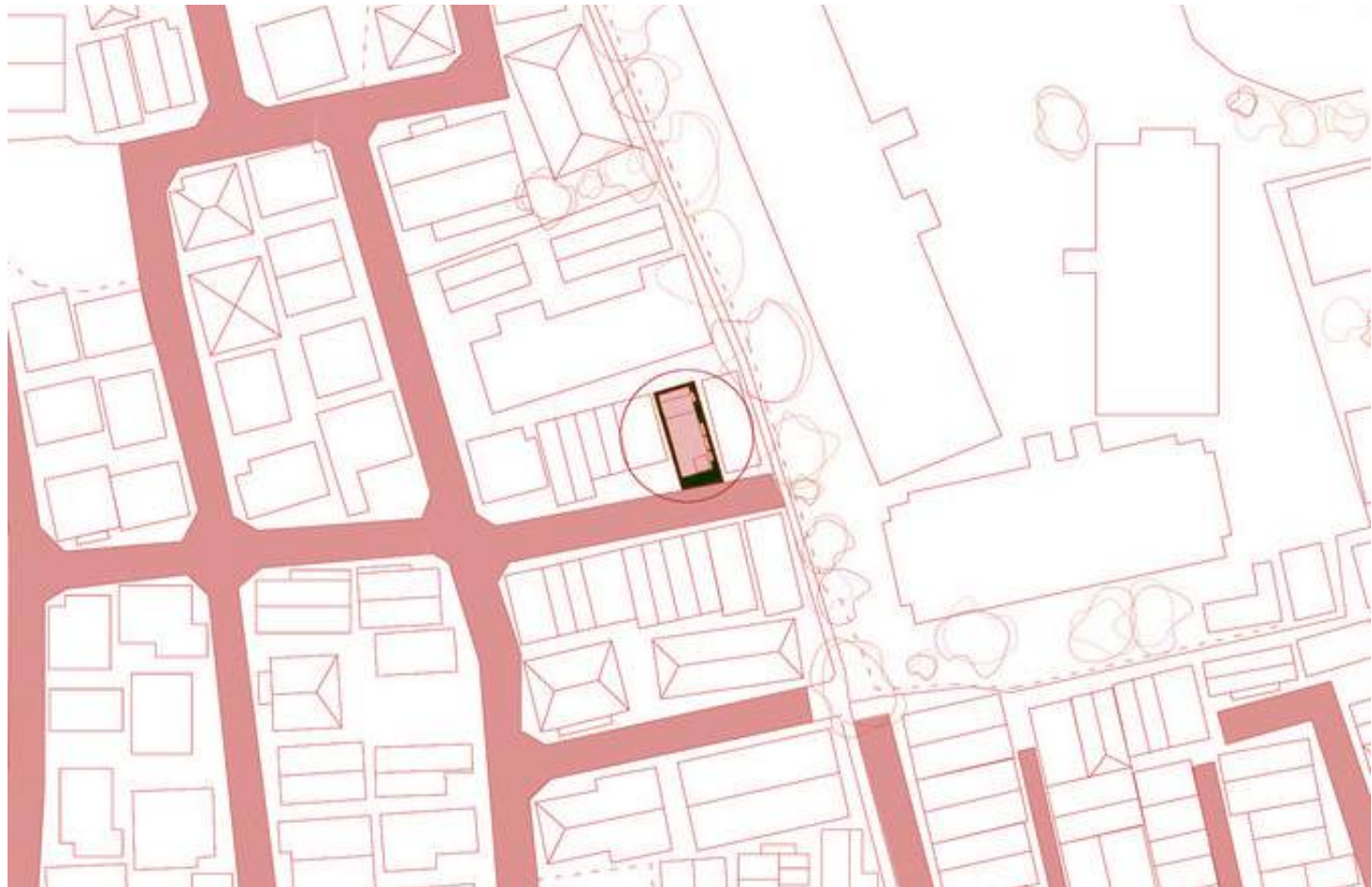
Due to road restrictions, vehicle access is limited. Thus, Tato Architects decided to construct this household using lightweight materials such as steel sections for columns and beams, and 75mm deck plates for the floors.

Compared to the surrounding houses, House in Itami does not have a front yard facing the streets. Additionally, the main entrance opens on the side.

Another interesting finding was that, instead of placing the more private spaces on the innermost part of the structure, the bedroom and the private bathroom are located on the first floor, making them the first spaces one would encounter upon entering the house.

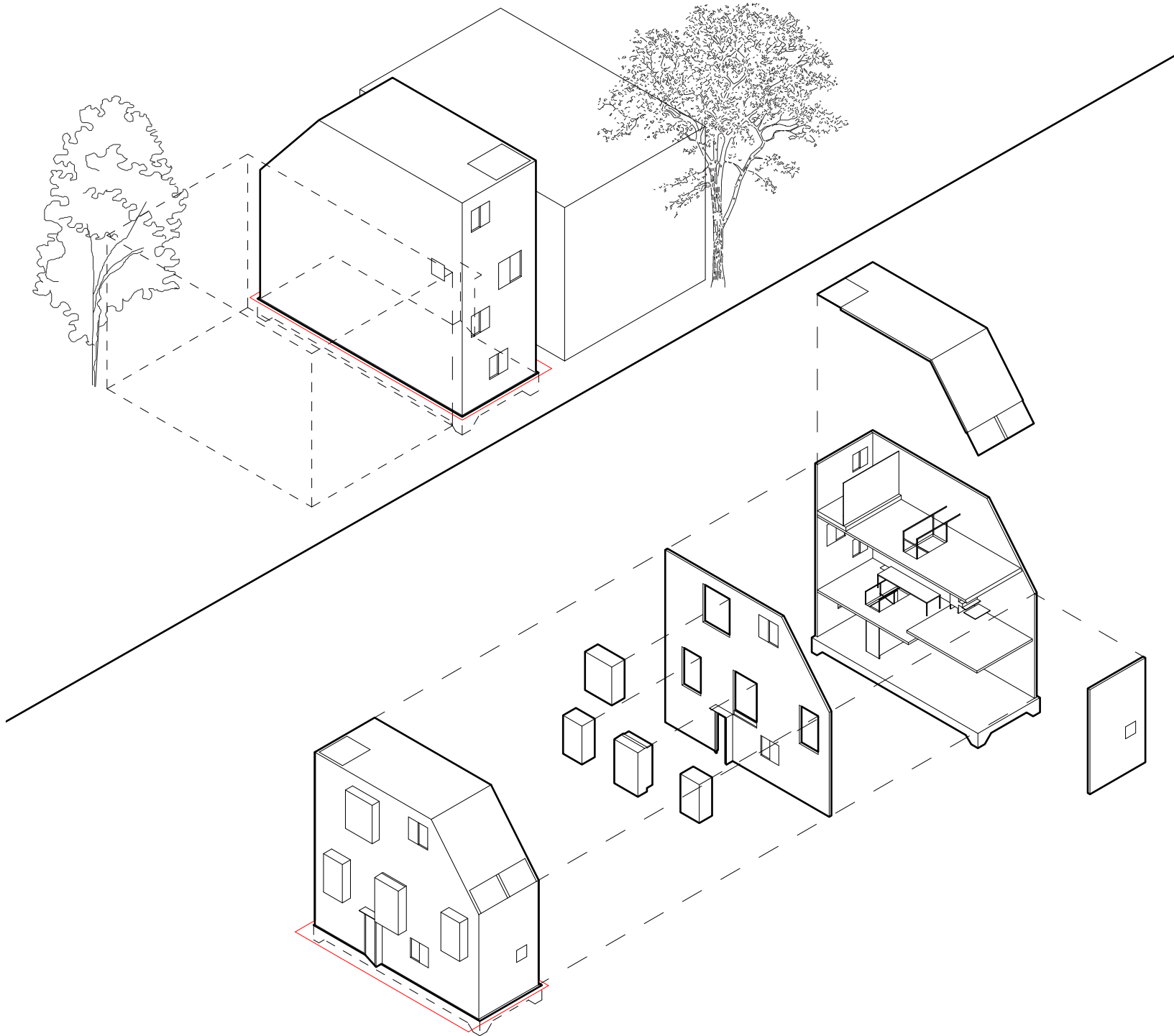
Considering that the covering area of the structure is very small, Tato decided to intergrate furnitures into architecture, while also making stairs, handrails as furnitures.

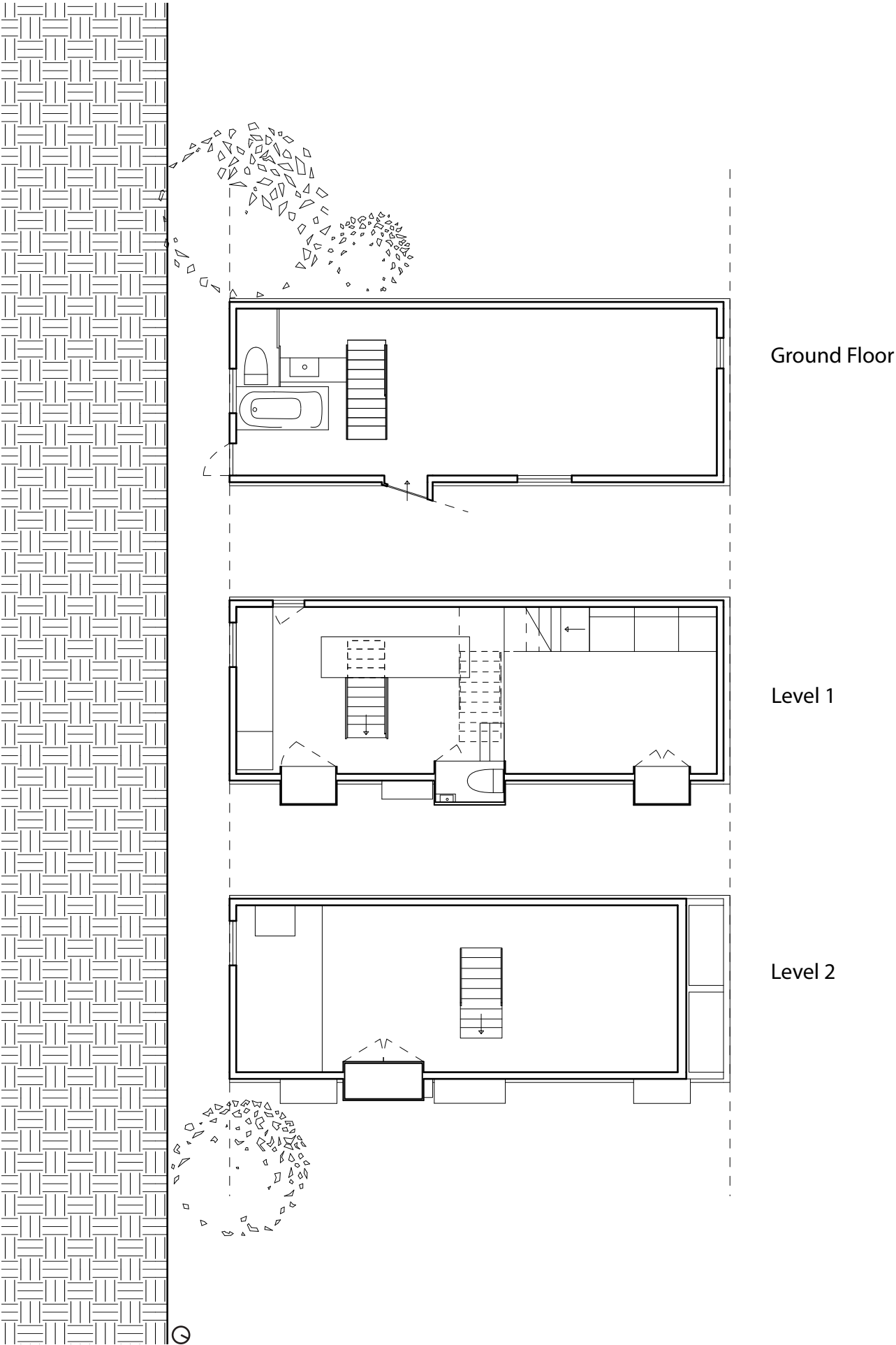
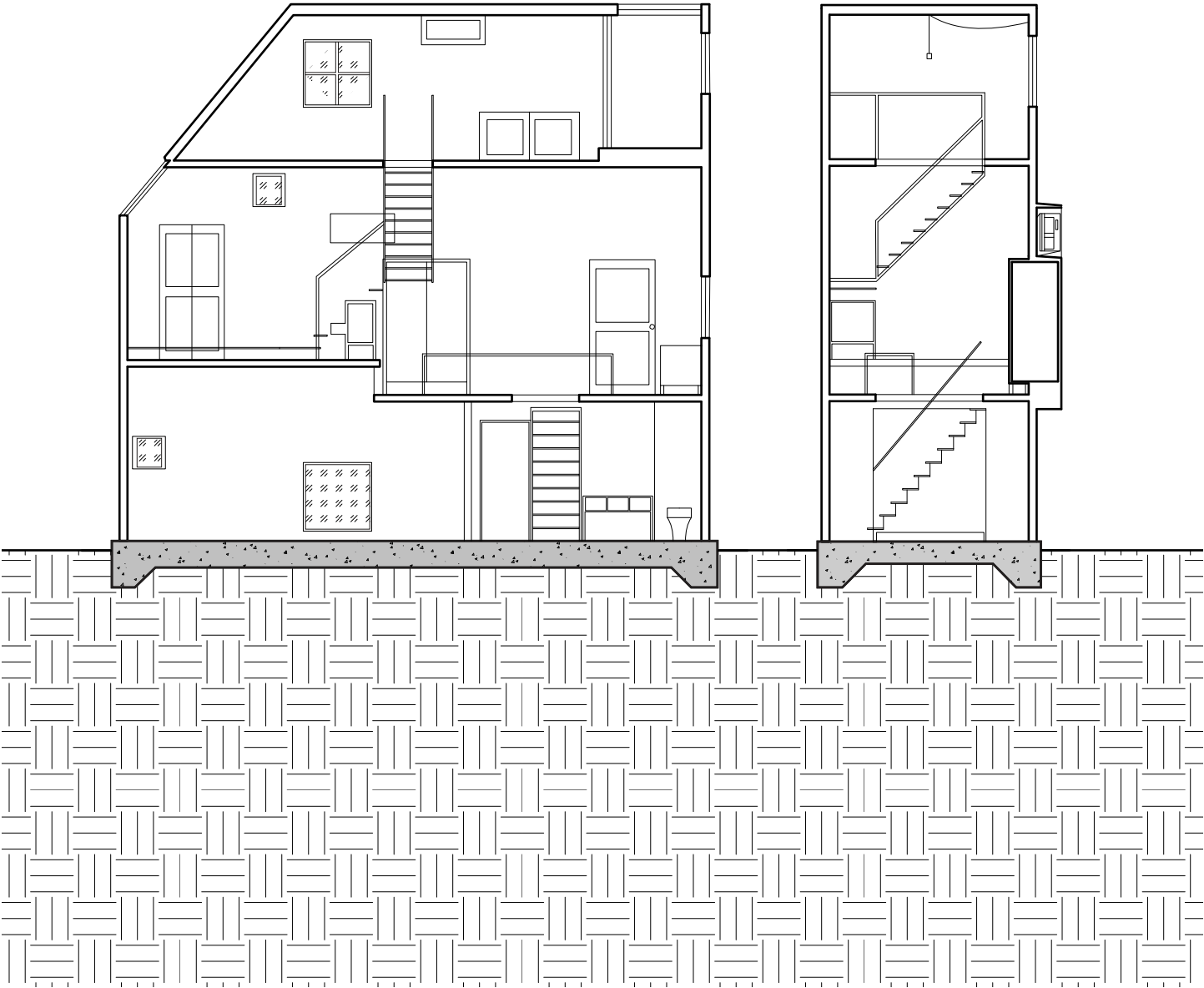
A problem that the architect faced with this location was the placement of the openings. Since the space is bounded by two other buildings, the architects chose to avoid openings that directly face the two other neighbors and created a level difference on the first floor, allowing natural light to pass through and reach the ground floor.



PRECEDENT DRAWINGS

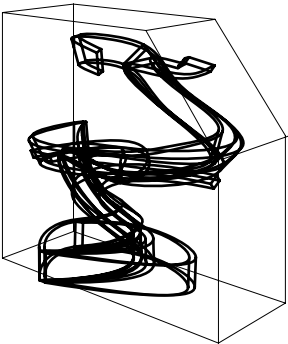
Precedent axonometric with contexts and exploded Axonometric



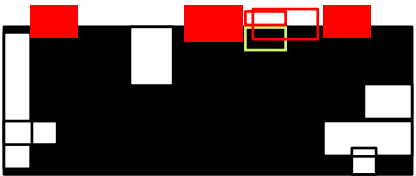


PRECEDENT DIAGRAMINGS

Precedent Diagrams - Circulation, +- Spaces, Common vs Private



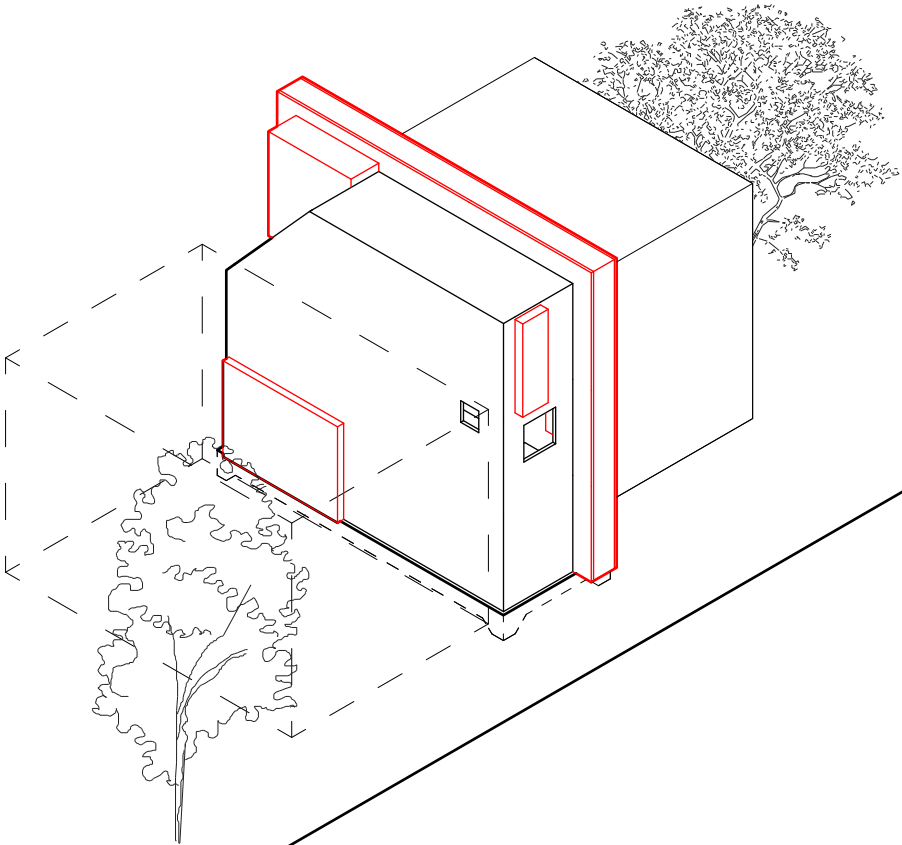
Circulation



+ - Spaces

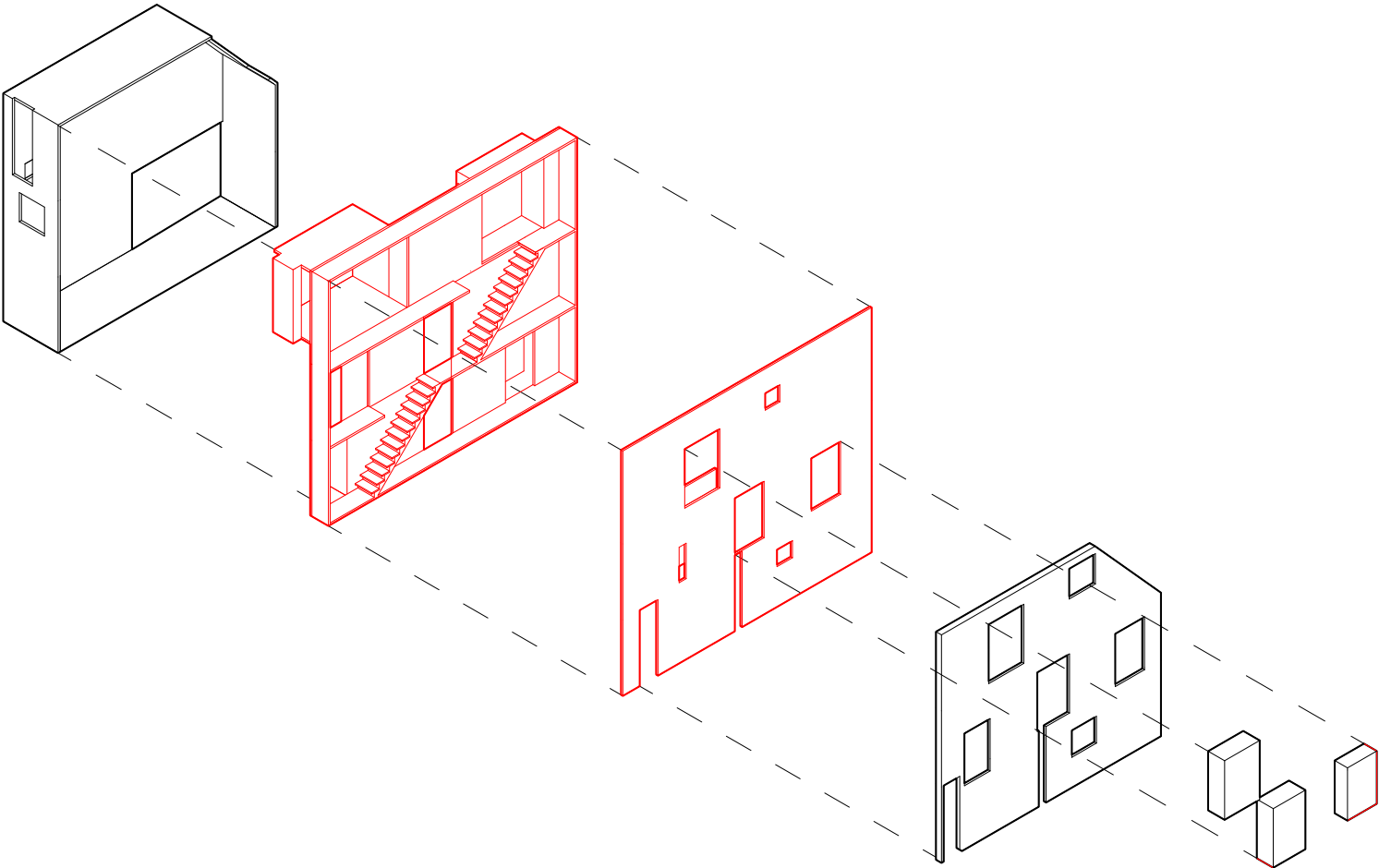


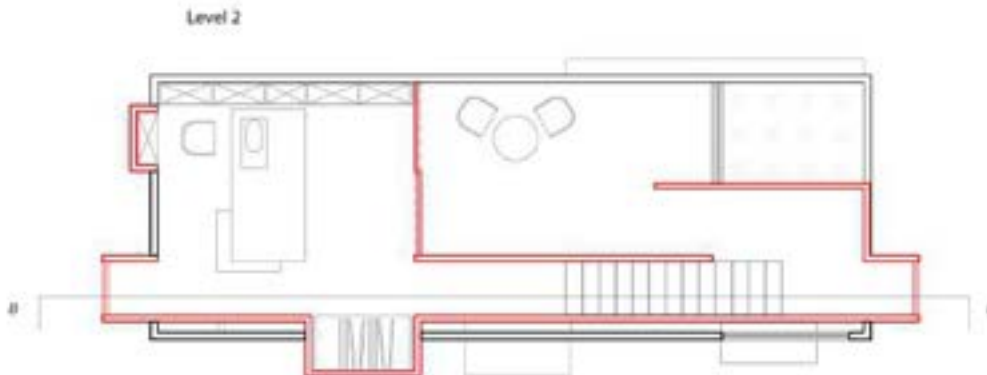
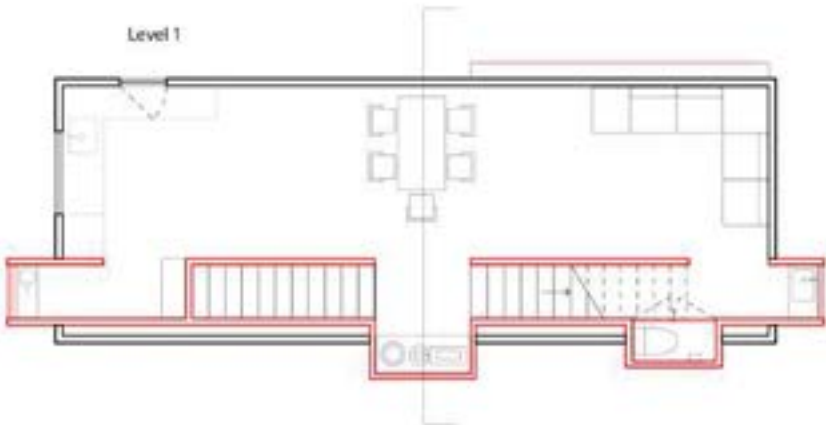
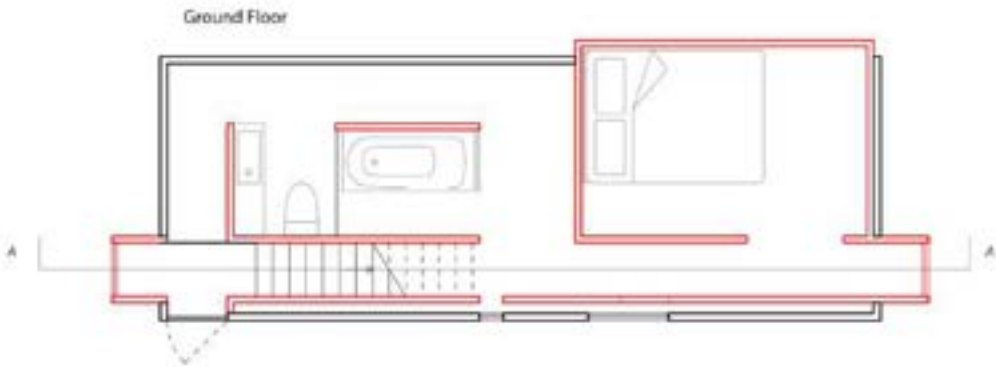
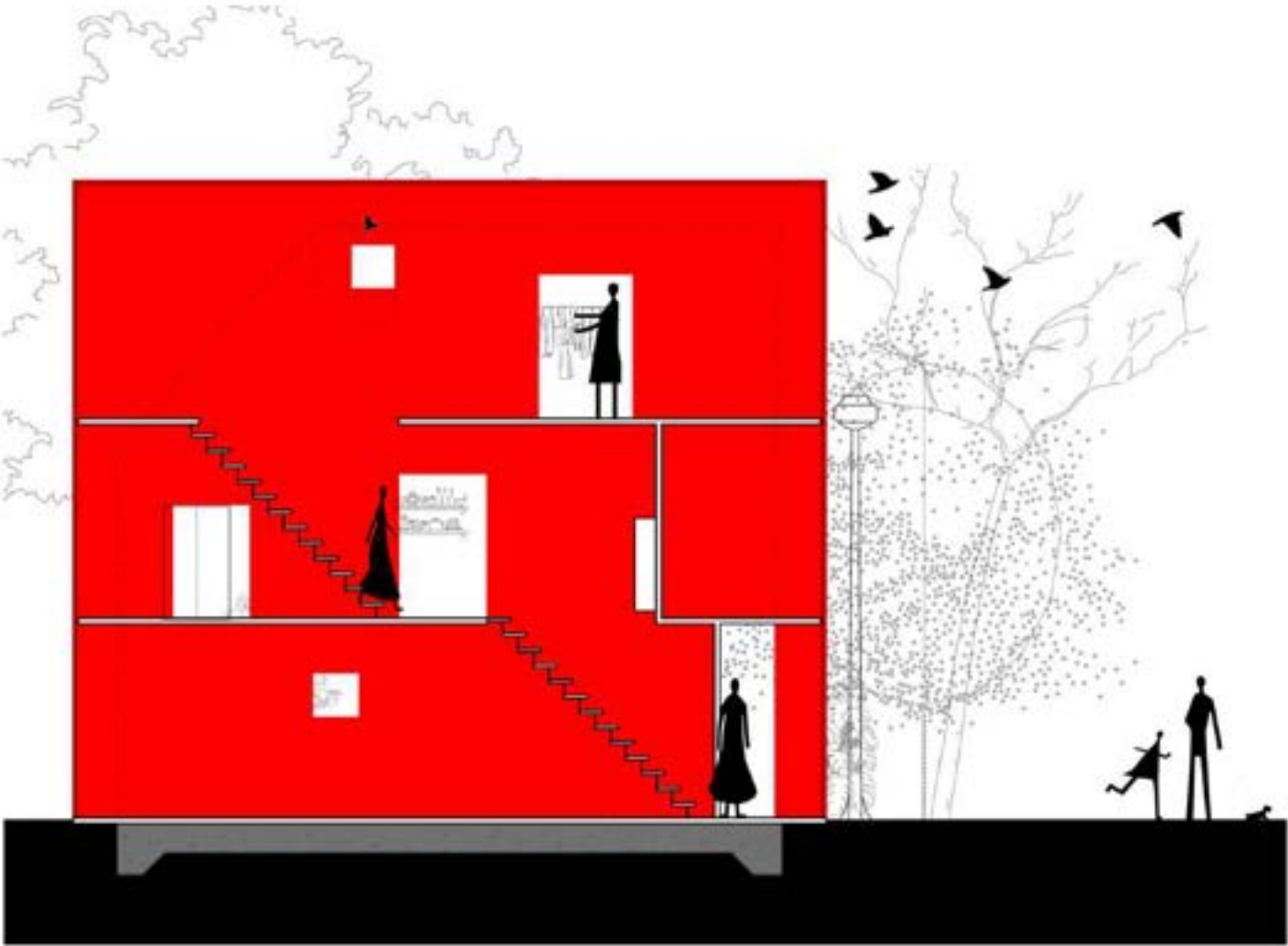
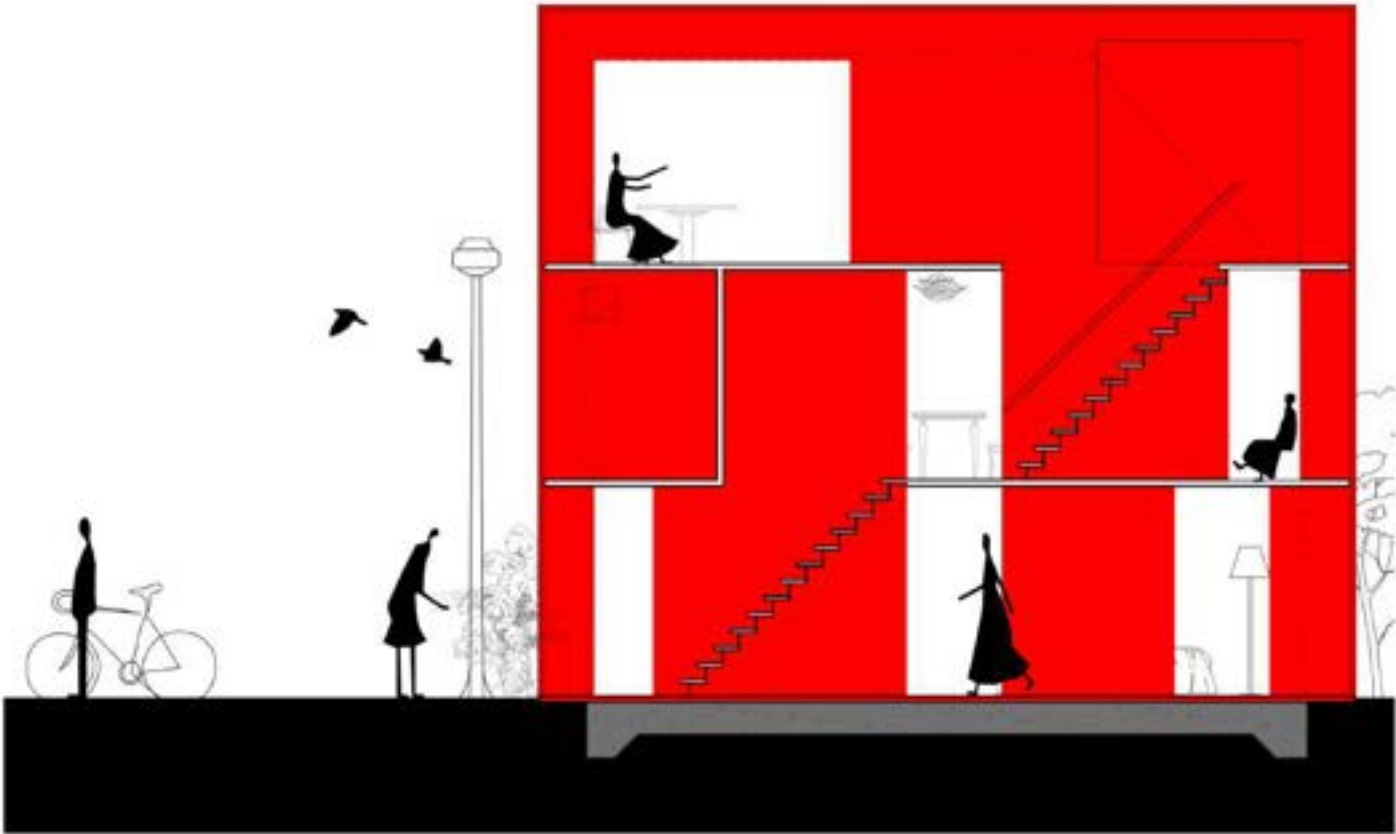
Common vs Private



DESIGN WITH INTERVENTION

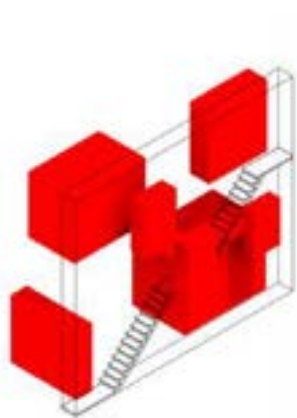
Intervention axonometric with contexts and exploded Axonometric



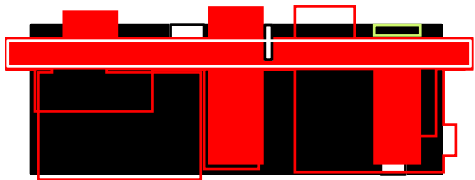


INTERVENTION DIAGRAMINGS

Intervention Diagrams - Circulation, +- Spaces, Common vs Private



Circulation

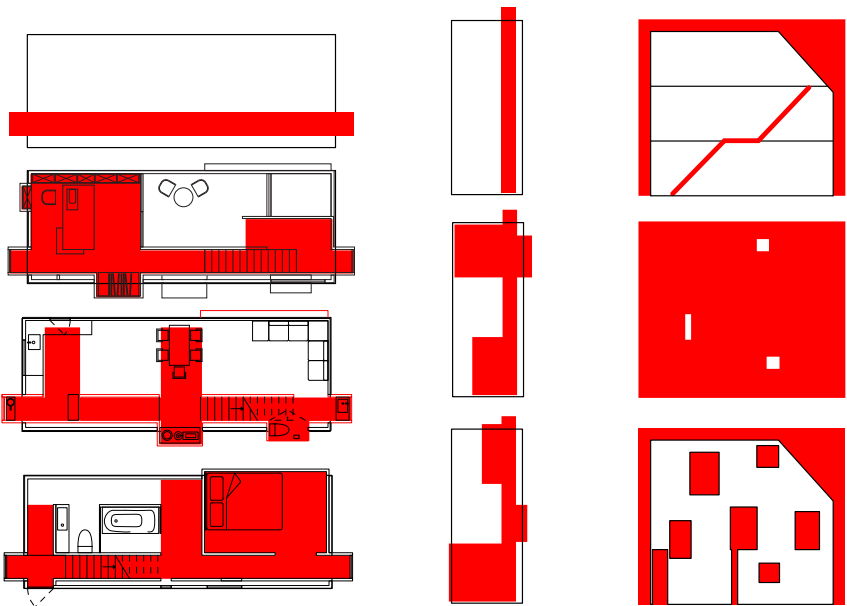
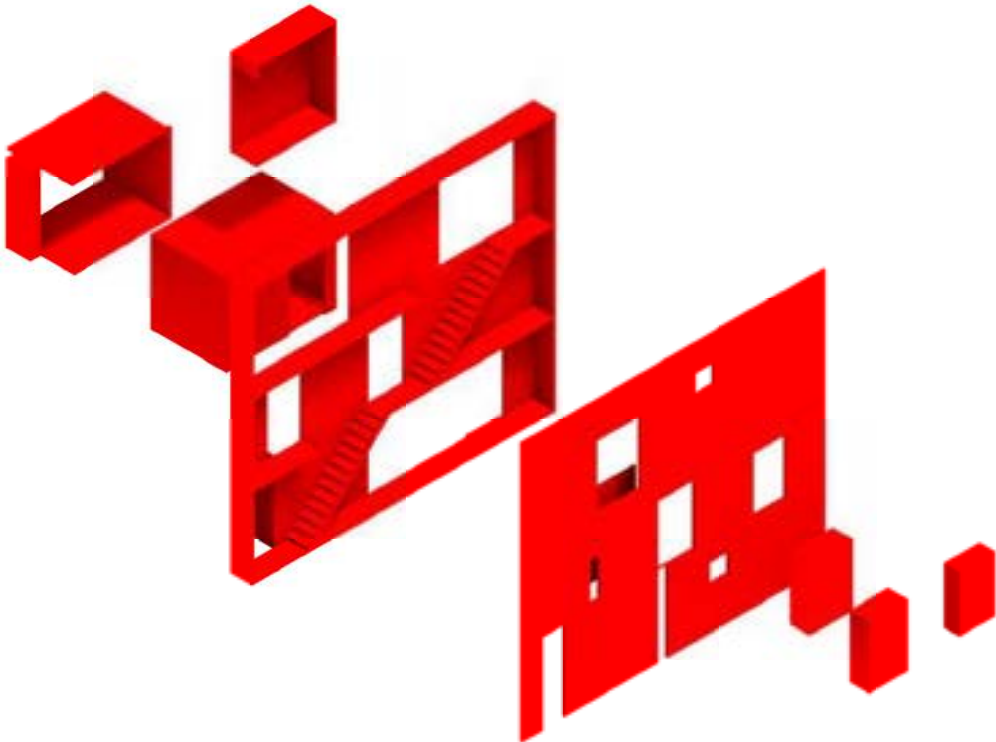
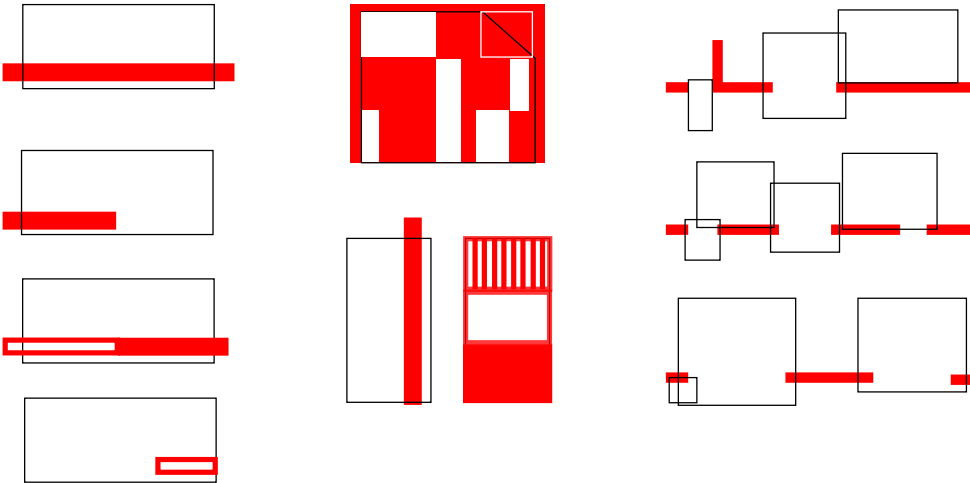


+ - Spaces



Common vs Private

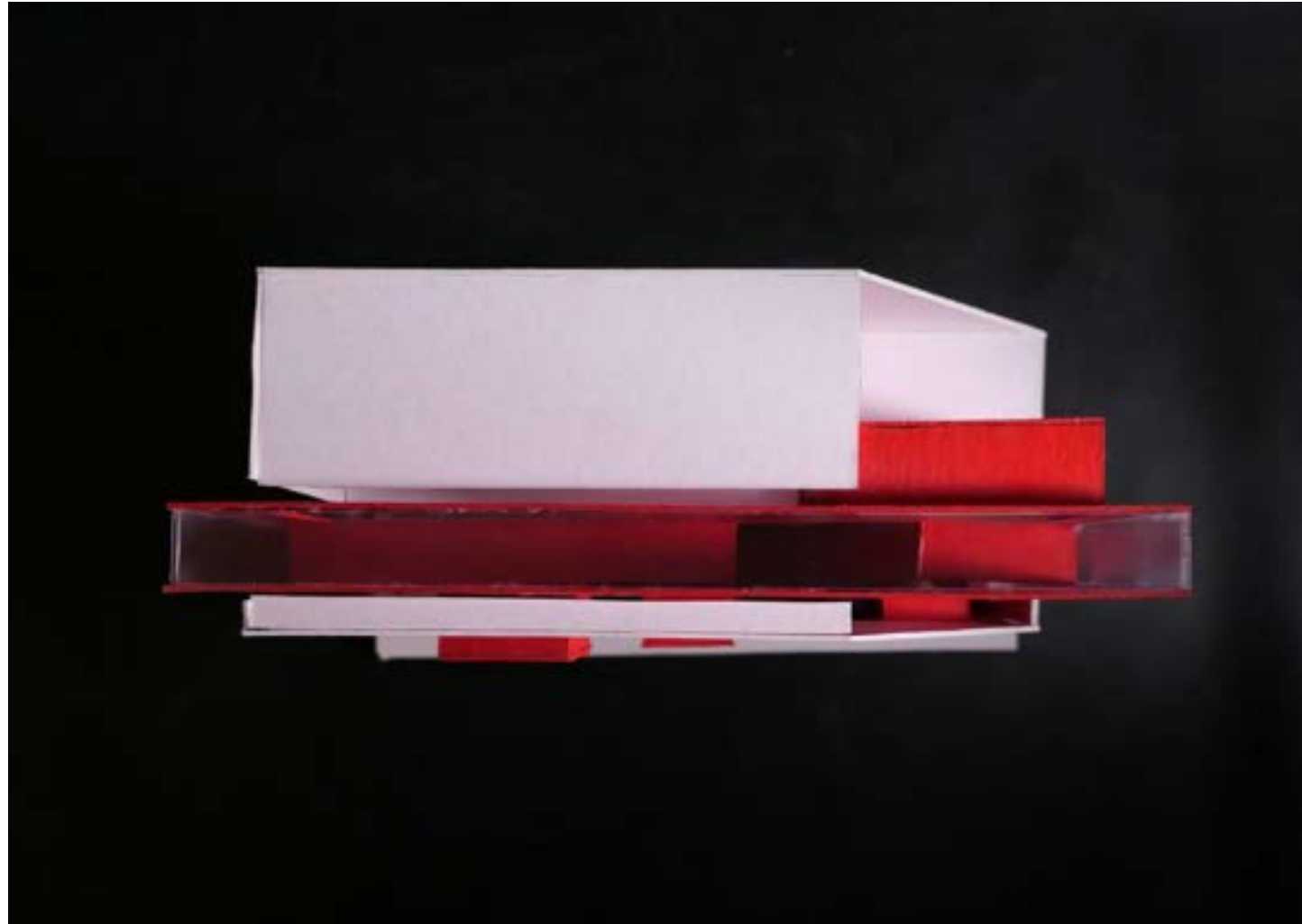
Intervention Diagrams - Spacial Organization, Exploded Axon, Intervention in Existing Structure

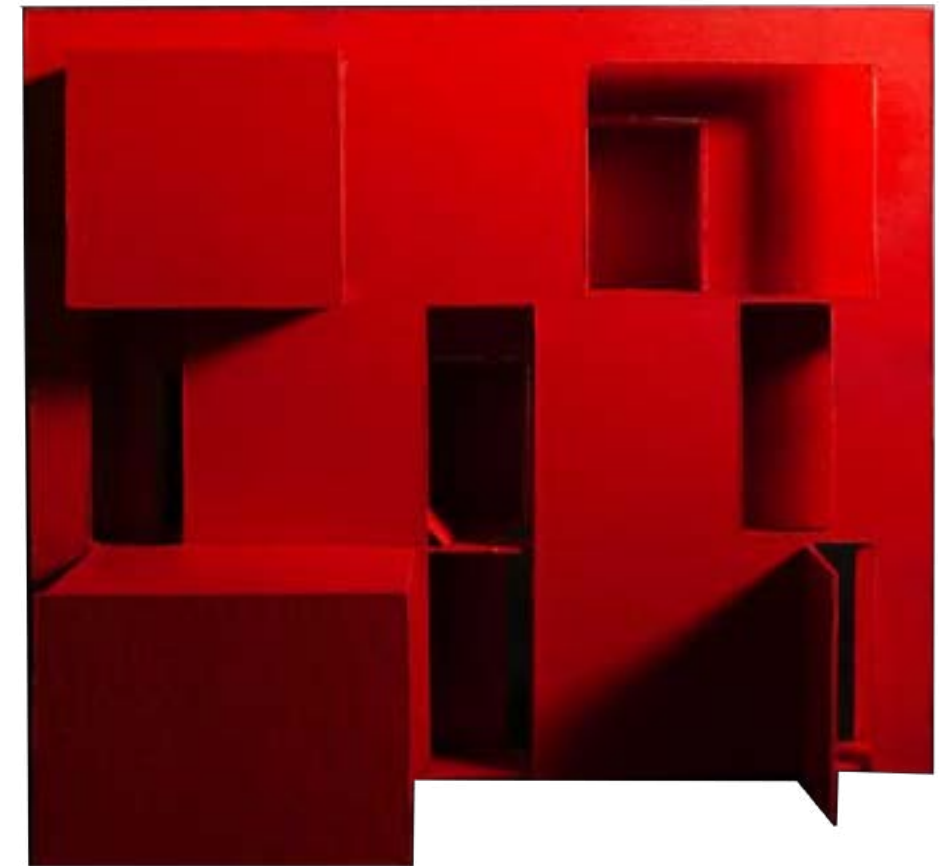


DESIGN WITH INTERVENTION MODEL

1 / 4" = 1'



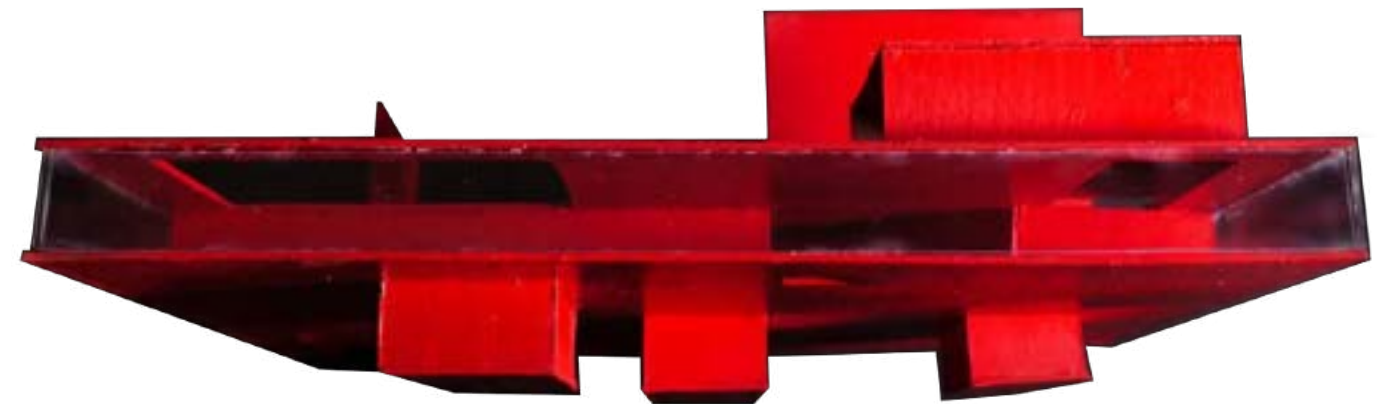




Intervention Model Side Elevation(Left)



Intervention Model Side Elevation(Right)



Intervention Model Top Elevation





